Module 2

BOOLEAN ALGEBRA.

BINARY VARIABLES.

The variables with values zeroes or ones represented by English alphabets A, B, c etc. The relationship b/n (rules for manipulations of binary variables) the binary variables is known as Boolean Algebra.

1. $A + 0 = A$.

   Proof : If $A = 0$, $0 + 0 = 0$

   If $A = 1$, $1 + 0 = 1$.

2. $A \cdot 0 = 0$

3. $A + 1 = 1$

4. $A \cdot 1 = A$

5. $A + A = A$.

6. $A \cdot A = A$.

7. $\overline{\overline{A}} = A$.

8. $A + \overline{A} = 1$

9. $A \cdot \overline{A} = 0$.

10. $\left.\begin{array}{l} A + B = B + A. \\ A \cdot B = B \cdot A. \end{array}\right\}$ Commutative Law.

11. $\left.\begin{array}{l} A \cdot (B + C) = AB + AC \\ A + BC = (A+B) \cdot (A+C) \end{array}\right\}$ Distributive Law.

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12. $A + (B+C) = (A+B) + C$
$A \cdot (B C) = (A \cdot B) \cdot C.$ } Associative Law

13. $A + AB = A.$

14. $A \cdot (A+B) = A.$

15. $A + \bar{A} B = A + B.$

16. $A(\bar{A} + B) = AB.$

17. $AB + A\bar{B} = A.$

18. $(A+B)(A+\bar{B}) = A.$

De' Morgan's Theorem.

19. $\overline{A+B} = \bar{A} \cdot \bar{B}.$

20 $\overline{A \cdot B} = \bar{A} + \bar{B}.$

Problems

Q.- $(A+B)(A+\bar{B})(\bar{A}+C).$

$= (A \cdot A + A\bar{B} + AB + 0)(\bar{A}+C)$

$= (A + A\bar{B} + AB)(\bar{A}+C) = A(1 + \bar{B} + AB)(\bar{A}+C)$

$= A(\bar{A}+C)$

$= \underline{\underline{AC}}$

Q.- $A\bar{B} + \bar{A}B + AB + \bar{A}\bar{B}$

$= A(\bar{B}+B) + \bar{A}(B+\bar{B}) = A + \bar{A} = \underline{\underline{1}}$

Q:- $AB C + A\bar{B}C + AB\bar{C}$

$= AC(B + \bar{B}) + (AB\bar{C})$

$= AC + AB\bar{C}$

$= A(C + B\bar{C})$

$= A(C + B)(C + \bar{C})$

$= A \cdot (B + C)$

Q:- $AB\bar{C} + A\bar{B}C + \bar{A}BC + ABC + A\bar{B}\bar{C}$

$= AB(C + \bar{C}) + A\bar{B}(C + \bar{C}) + \bar{A}BC.$

$= AB + A\bar{B} + \bar{A}BC$

$= A(B + \bar{B}) + \bar{A}BC$

$= A + \bar{A}BC = (A + \bar{A})(A + BC)$

$= A + BC$

Q:- $A + \bar{A}B + AB$

$= A + B(A + \bar{A})$

$= A + B$

Q.- $\bar{A}\bar{C} + B\bar{C} + \bar{A}BC + ABC$

$= \bar{A}\bar{C} + B\bar{C} + BC(A + \bar{A})$

$= \bar{A}\bar{C} + B\bar{C} + BC$

$= \bar{A}\bar{C} + B(\bar{C} + C)$

$= \bar{A}\bar{C} + B$

Q:- $A + \bar{B}C(A + \bar{B}C)$

$= A + A\bar{B}C + \bar{B}C.$

$= A + \bar{B}C(A+1)$

$= \underline{\underline{A + \bar{B}C}}$

Q:- $\overline{\overline{A+B} + \bar{C}}$

$= \overline{\overline{A+B}} \cdot \overline{\bar{C}}$

$= \underline{\underline{(A+B) \cdot C}}$

Q:- $Y = A(B + C(\overline{AB + AC}))$

$= A[B + C(\overline{AB} \cdot \overline{AC})]$

$= A(B + C[(\bar{A} + \bar{B})(\bar{A} + \bar{C})])$

$= A[B + C(\bar{A} + \bar{A}\bar{C} + \bar{A}\bar{B} + \bar{B}\bar{C})]$

$= A[B + C\bar{A} + \bar{A}\bar{B}C]$

$= \underline{\underline{AB}}$

Q:- P.T $ab + bc + \bar{a}c = ab + \bar{a}c$

LHS $= ab + bc + \bar{a}c$

$= ab + (a + \bar{a})bc + \bar{a}c$

$= ab + abc + \bar{a}bc + \bar{a}c$

$= ab(1 + c) + \bar{a}c(b + 1)$

$= ab + \bar{a}c = \text{RHS}$

Q:- $abc\,(ab+\bar{c})\,(bc+ac)$

$= (abc+0)(\;bc+ac)$

$= abc + abc$

$= \underline{\underline{abc}}$

Q:- P.T, $A\bar{B}\bar{D} + AB\bar{C}\bar{D} + ABC\bar{D} + \bar{A}\bar{B}\bar{D} + \bar{A}BC\bar{D}$

$= \bar{B}\bar{D} + A\bar{D} + C\bar{D}$.

LHS $= A\bar{B}\bar{D} + \underline{AB\bar{C}\bar{D}} + \underline{ABC\bar{D}} + \bar{A}\bar{B}\bar{D} + \bar{A}BC\bar{D}$

$= (A+\bar{A})\,\bar{B}\bar{D} + AB\bar{D}\,(C+\bar{C}) + \bar{A}BC\bar{D}$.

$= \bar{B}\bar{D} + AB\bar{D} + \bar{A}BC\bar{D}$

$= \bar{B}\bar{D} + B\bar{D}\,(A + \bar{A}C)$

$= \bar{B}\bar{D} + B\bar{D}\,(A+\bar{A})\,(A+C)$

$= \bar{B}\bar{D} + B\bar{D}\,(A+C)$

$= \bar{B}\bar{D} + AB\bar{D} + BC\bar{D}$

$= \bar{D}\,(\bar{B} + AB + BC)$

$= \bar{D}\,((\bar{B}+B)\,(\bar{B}+A) + BC)$

$= \bar{D}\,(\bar{B} + A + BC)$

$= \bar{D}\,(A + (\;+\bar{B})(\bar{B}+C))$

$= \bar{D}\,(A + \bar{B}+C)$

$= \underline{\underline{A\bar{D} + \bar{B}\bar{D} + C\bar{D}}}$

Q:- $(a+b)(b+c)(\bar{a}+c) = (a+b)(\bar{a}+c)$

$$LHS = (a+b)(b+c)(\bar{a}+c).$$

$$= (ab + ac + b + bc)\cdot(\bar{a} + c)$$

$$= \underline{abc} + ac + \bar{a}b + bc + \underline{\bar{a}bc} + bc$$

$$= bc(a+\bar{a}) + ac + \bar{a}b + bc$$

$$= \underline{\underline{bc + ac + \bar{a}b.}}$$

$$RHS = (a+b)(\bar{a}+c)$$

$$= ac + a\bar{b} + \underline{\underline{bc}} = LHS$$

Q:- $(\bar{w}+x)y + wxy$

$$= \bar{w}y + xy + wxy$$

$$= \bar{w}y + xy(1+w)$$

$$= \bar{w}y + xy$$

$$= \underline{\underline{y(\bar{w}+x)}}$$

Q:- $\overline{(a+\bar{b})(\bar{c}+d)}$

$$= \overline{a+\bar{b}} + \overline{\bar{c}+d}$$

$$= \bar{a}\cdot\bar{\bar{b}} + \bar{\bar{c}}\cdot\bar{d}$$

$$= \underline{\underline{\bar{a}\cdot b + c\bar{d}}}$$

Q: $\overline{w\bar{x}(y+\bar{z})}$

$$= \overline{w\bar{x}} + \overline{y+\bar{z}} = \bar{w} + \bar{\bar{x}} + \bar{y} + \bar{\bar{z}}$$

$$= \underline{\underline{\bar{w} + x + \bar{y} + z}}$$

Standard Representation for Logical Functions.

There are two types of Std. representations

(1) Sum of product (SOP)

eg:- $Y = AB + B\bar{C} + \bar{A}\bar{C}$.

(2) Product of sum (POS)

eg:- $Y = (\bar{A} + \bar{B})(A + C)(A + B)$.

$Y = (A + BC)(B + A\bar{C})$.

SOP Form

$$Y = AB + A\bar{C} + BC.$$

POS Form

$$Y = (A + BC)(B + A\bar{C})$$
$$= (A + B)(A + C)(A + B)(B + \bar{C})$$
$$= (A + B)(A + C)(B + \bar{C})$$

Std SOP

$$Y = AB + A\bar{C} + BC.$$
$$= AB(C + \bar{C}) + A\bar{C}(B + \bar{B}) + BC(A + \bar{A})$$
$$= ABC + AB\bar{C} + AB\bar{C} + A\bar{B}\bar{C} + ABC + \bar{A}BC$$
$$= ABC + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC$$

Std POS

$$Y = (A+B)(A+C)(B+\bar{C})$$

$$= (A+B+C\bar{C})(A+C+B\bar{B})(B+\bar{C}+A\bar{A})$$

$$= (A+B+C)(A+B+\bar{C})(A+B+C)(A+\bar{B}+C)$$

$$(A+B+\bar{C})(\bar{A}+B+\bar{C})$$

$$= (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+\bar{C})$$

				SOP	POS
0	0	0	0	$\bar{A}\bar{B}\bar{C}$ — m_0	$A+B+C$ — M_0
1	0	0	1	$\bar{A}\bar{B}C$ — m_1	$A+B+\bar{C}$ — M_1
2	0	1	0	$\bar{A}B\bar{C}$ — m_2	$A+\bar{B}+C$ — M_2
3	0	1	1	$\bar{A}BC$ — m_3	$A+\bar{B}+\bar{C}$ — M_3
4	1	0	0	$A\bar{B}\bar{C}$ — m_4	$\bar{A}+B+C$ — M_4
5	1	0	1	$A\bar{B}C$ — m_5	$\bar{A}+B+\bar{C}$ — M_5
6	1	1	0	$AB\bar{C}$ — m_6	$\bar{A}+\bar{B}+C$ — M_6
7	1	1	1.	ABC — m_7	$\bar{A}+\bar{B}+\bar{C}$ — M_7.
				Min-terms	Max-terms

$$Y = \bar{A}\bar{B}\bar{C} + AB\bar{C} + ABC$$

$$= 000 + 110 + 111$$

$$= m_0 + m_6 + m_7 = \sum m(0,6,7).$$

$$Y = (\underset{0}{A}+\underset{0}{B}+\underset{0}{C})(\underset{1}{\bar{A}}+\underset{1}{\bar{B}}+\underset{0}{C})(\underset{1}{\bar{A}}+\underset{1}{\bar{B}}+\underset{1}{\bar{C}})$$

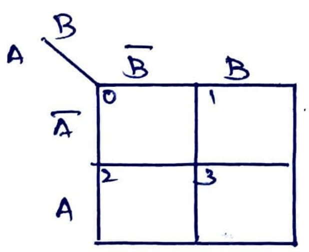
$$= M_0 \cdot M_6 \cdot M_7$$

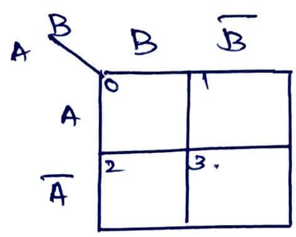
$$= \pi M(0,6,7)$$

Decimal	Binary				Min-Terms	Max-Terms
0	0	0	0	0	$\bar{A}\bar{B}\bar{C}\bar{D}$ — m_0	$A+B+C+D$ — M_0
1	0	0	0	1	$\bar{A}\bar{B}\bar{C}D$ — m_1	$A+B+C+\bar{D}$ — M_1
2	0	0	1	0	$\bar{A}\bar{B}C\bar{D}$ — m_2	$A+B+\bar{C}+D$ — M_2
3	0	0	1	1	$\bar{A}\bar{B}CD$ — m_3	$A+B+\bar{C}+\bar{D}$ — M_3
4	0	1	0	0	$\bar{A}B\bar{C}\bar{D}$ — m_4	$A+\bar{B}+C+D$ — M_4
5	0	1	0	1	$\bar{A}B\bar{C}D$ — m_5	$A+\bar{B}+C+\bar{D}$ — M_5
6	0	1	1	0	$\bar{A}BC\bar{D}$ — m_6	$A+\bar{B}+\bar{C}+D$ — M_6
7	0	1	1	1	$\bar{A}BCD$ — m_7	$A+\bar{B}+\bar{C}+\bar{D}$ — M_7
8	1	1	0	0	$AB\bar{C}\bar{D}$ — m_8	$\bar{A}+\bar{B}+C+D$ — M_8
9	1	0	0	1	$A\bar{B}\bar{C}D$ — m_9	$\bar{A}+B+C+\bar{D}$ — M_9
10	1	0	1	0	$A\bar{B}C\bar{D}$ — m_{10}	$\bar{A}+B+\bar{C}+D$ — M_{10}
11	1	0	1	1	$A\bar{B}CD$ — m_{11}	$\bar{A}+B+\bar{C}+\bar{D}$ — M_{11}
12	1	0	0	0	$A\bar{B}\bar{C}\bar{D}$ — m_{12}	$\bar{A}+B+C+D$ — M_{12}
13	1	1	0	1	$AB\bar{C}D$ — m_{13}	$\bar{A}+\bar{B}+C+\bar{D}$ — M_{13}
14	1	1	1	0	$ABC\bar{D}$ — m_{14}	$\bar{A}+\bar{B}+\bar{C}+D$ — M_{14}
15	1	1	1	1	$ABCD$ — m_{15}	$\bar{A}+\bar{B}+\bar{C}+\bar{D}$ — M_{15}

KARNAUGH MAP / K-MAP.

2 Variable. (A, B).





3-Variable

AB \ C	$\bar{C}$ 0	C 1
$\bar{A}\bar{B}$ 00	0	1
$\bar{A}B$ 01	2	3
AB 11	6	7
$A\bar{B}$ 10	4	5

AB \ C	C 0	$\bar{C}$ 1
$A+B$ 00	0	1
$A+\bar{B}$ 01	2	3
$\bar{A}+\bar{B}$ 11	6	7
$\bar{A}+B$ 10	4	5

4-Variable

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0	1	3	2
$\bar{A}B$ 01	4	5	7	6
AB 11	12	13	15	14
$A\bar{B}$ 10	8	9	11	10

AB \ CD	$C+D$ 00	$C+\bar{D}$ 01	$\bar{C}+\bar{D}$ 11	$\bar{C}+D$ 10
$A+B$ 00	0	1	3	2
$A+\bar{B}$ 01	4	5	7	6
$\bar{A}+\bar{B}$ 11	12	13	15	14
$\bar{A}+B$ 10	8	9	11	10

Reduction of Boolean Expressions using K-MAP.

Q:- $Y = \sum m(4,6,7)$

AB \ C	$\bar{C}$ 0	C 1
$\bar{A}\bar{B}$ 00	$_0$	$_1$
$\bar{A}B$ 01	$_2$	$_3$
✓ AB 11	1_6	1_7
✓ $A\bar{B}$ 10	1_4	$_5$

$Y = AB + A\bar{C}$

$2^0 = 1$
$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16$

$Y = \sum m(0,1,4,5,7)$

AB \ C	$\bar{C}$ 0	C 1
$\bar{A}\bar{B}$ 00	1_0	1_1
$\bar{A}B$ 01	$_2$	$_3$
AB 11	$_6$	1_7
$A\bar{B}$ 10	1_4	1_5

$Y = \bar{B} + AC$

Q:- $Y = \sum m(0,1,2,3,7,8,9,10,11,12,13)$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	1_0	1_1	1_3	1_2
$\bar{A}B$ 01	$_4$	$_5$	1_7	$_6$
AB 11	1_{12}	1_{13}	$_{15}$	$_{14}$
$A\bar{B}$ 10	1_8	1_9	1_{11}	1_{10}

$Y = \bar{B} + A\bar{C} + \bar{A}CD$

$$Y = f(A,B,C,D) = \Sigma m(0,1,2,3,5,7,8,9,11,14)$$

AB \ CD	$\bar C \bar D$ 00	$\bar C D$ 01	CD 11	$C\bar D$ 10
$\bar A \bar B$ 00	1 (0)	1 (1)	1 (3)	1 (2)
$\bar A B$ 01	(4)	1 (5)	1 (7)	(6)
AB 11	(12)	(13)	(15)	1 (14)
$A\bar B$ 10	1 (8)	1 (9)	1 (11)	(10)

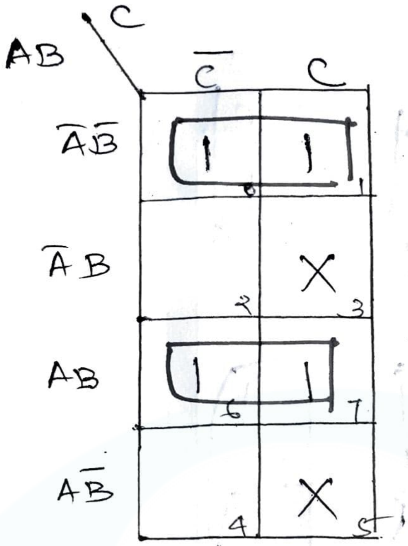
$$Y = \bar A \bar B + \bar B \bar C + \bar B D + \bar A . D + A B C \bar D$$

$$Y = f(A,B,C,D) = \pi M(0,1,3,5,6,7,10,14,15)$$

AB \ CD	$C+D$ 00	$C+\bar D$ 01	$\bar C+\bar D$ 11	$\bar C+D$ 10
$A+B$ 00	0 (0)	0 (1)	0 (3)	(2)
$A+\bar B$ 01	(4)	0 (5)	0 (7)	0 (6)
$\bar A+\bar B$ 11	(12)	(13)	0 (15)	0 (14)
$\bar A+B$ 10	(8)	(9)	(11)	0 (10)

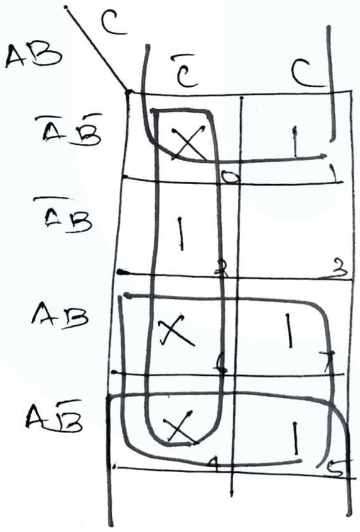
$$Y = (A+\bar D) \cdot (\bar B + \bar C) \cdot (A+B+C)(\bar A + \bar C + D)$$

$$Y = \sum m(0,1,6,7) + \sum d(3,5)$$



$$Y = AB + \bar{A}\bar{B}$$

$$Y = \sum m(1,2,5,7) + \sum d(0,4,6)$$



$$Y = \bar{C} + \bar{B} + A$$

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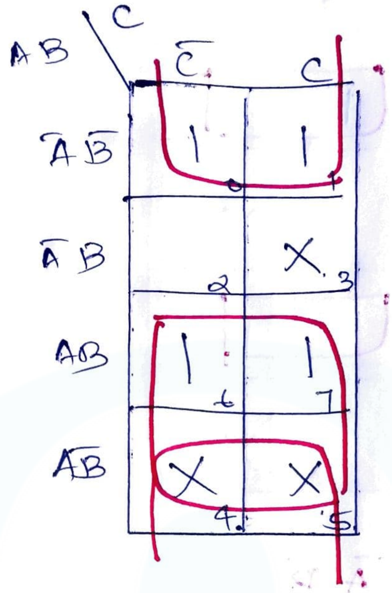
$Y = \Sigma m(0,1,6,7) + \Sigma d(3,4,5)$



$Y = A + \bar{B}$.

$Y = f(A,B,C,D) = \sum m(1,3,7,11,15) + d(0,2,5)$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	X (0)	1 (1)	1 (3)	X (2)
$\bar{A}B$	(4)	X (5)	1 (7)	(6)
AB	(12)	(13)	1 (15)	(14)
$A\bar{B}$	(8)	(9)	1 (11)	(10)

$Y = CD + \bar{A}D.$

$Y = \sum m(0,1,2,3,8,9,10,11,15) + d(4,5,14)$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 (0)	1 (1)	1 (3)	1 (2)
$\bar{A}B$	X (4)	X (5)	(7)	(6)
AB	(12)	(13)	1 (15)	X (14)
$A\bar{B}$	1 (8)	1 (9)	1 (11)	1 (10)

$Y = \bar{B} + AC.$

$$Y = \sum m(1, 2, 8, 9, 10, 14) + d(0, 3, 6, 11)$$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	X	1	X	1
$\bar{A}B$				X
AB				1
$A\bar{B}$	1	1	X	1

$$Y = \bar{B} + C\bar{D}$$

$$Y = \prod M(0, 1, 6, 7, 9, 10, 14) \cdot d(2, 3, 8)$$

AB \ CD

	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$
$A+B$	0	0	X	X
$A+\bar{B}$			0	0
$\bar{A}+\bar{B}$				0
$\bar{A}+B$	X	0		0

$$Y = (\bar{C}+D)(B+C)(A+\bar{C})$$

LOGIC GATES.

NOT, AND, OR, NOR, NAND, X-OR, X-NOR.

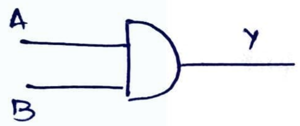
$\underbrace{\text{NOT, AND, OR}}_{\text{Fundamental Gates}}$, $\underbrace{\text{NOR, NAND}}_{\text{Universal gates}}$, X-OR, X-NOR.

(1) NOT.

$Y = \overline{A}$

A	Y
0	1
1	0

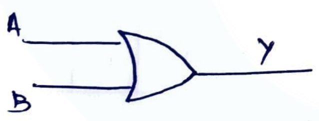
(2) AND



$Y = A \cdot B$.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

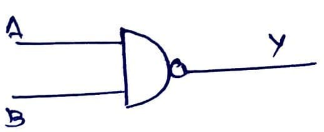
(3) OR



$Y = A + B$.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

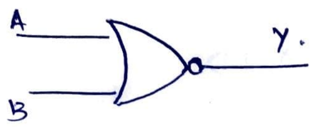
(4) NAND



$Y = \overline{A \cdot B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

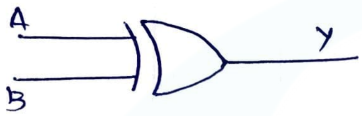
5. NOR



$Y = \overline{A+B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6. X-OR / EX-OR

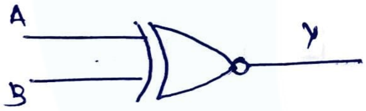


$Y = A \oplus B$

$= \bar{A}B + A\bar{B}$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7. X-NOR/EX-NOR



$Y = A \odot B = \overline{A \oplus B}$

$= \bar{A}\bar{B} + AB.$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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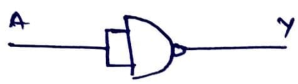
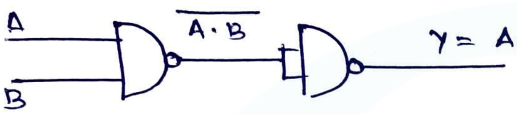
NAND

1. NOT



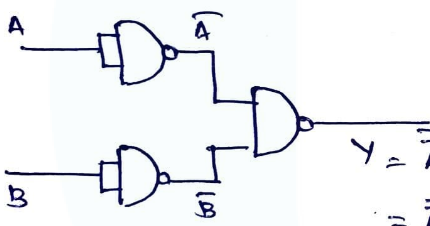
$Y = \overline{A \cdot A} = \overline{A}.$

2. AND



$Y = A \cdot B$

3. OR



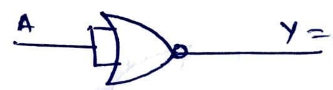
$Y = \overline{\overline{A} \cdot \overline{B}}$

$= \overline{\overline{A}} + \overline{\overline{B}}$

$= A + B$

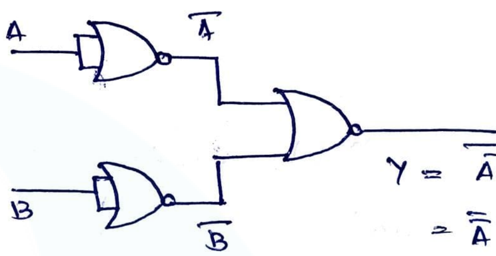
NOR.

NOT



$Y = \overline{A + A} = \overline{A}$

AND

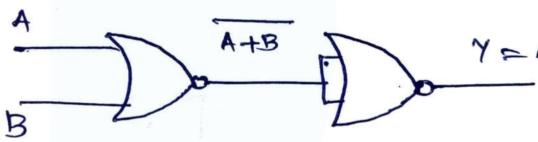


$Y = \overline{\overline{A} + \overline{B}}$

$= \overline{\overline{A}} \cdot \overline{\overline{B}}$

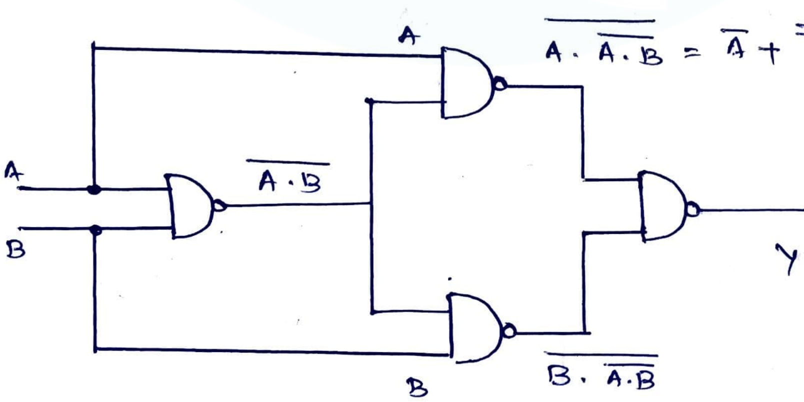
$= A \cdot B$

OR



$Y = A + B.$

4. X-OR using NAND



$\overline{A \cdot \overline{A \cdot B}} = \overline{A} + \overline{\overline{AB}} = \overline{A} + AB$

$\overline{B \cdot \overline{A \cdot B}}$

$= \overline{B} + \overline{\overline{A \cdot B}}$

$= \overline{B} + AB.$

$Y = \overline{(\overline{A} + AB)(\overline{B} + A \cdot B)}$

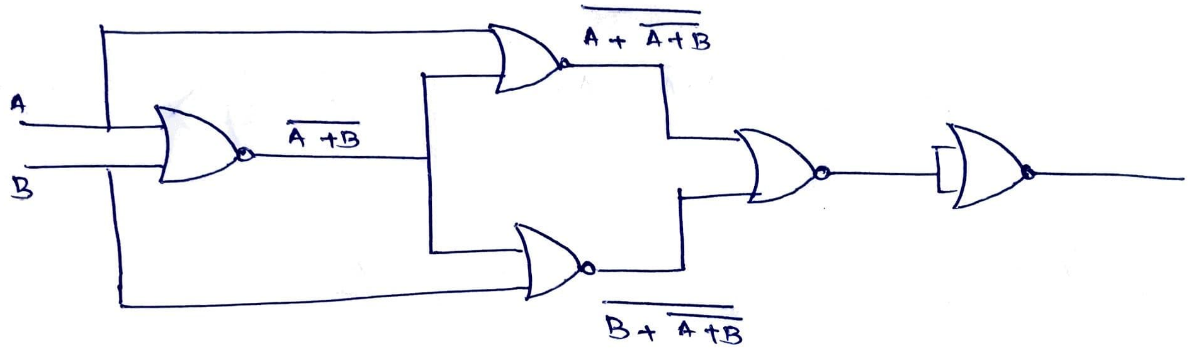
$= (A \cdot \overline{AB}) + (B \cdot \overline{A \cdot B})$

$= (A \cdot (\overline{A} + \overline{B})) + (B \cdot (\overline{A} + \overline{B}))$

$= A\overline{B} + \overline{A}B$

$= A \oplus B$

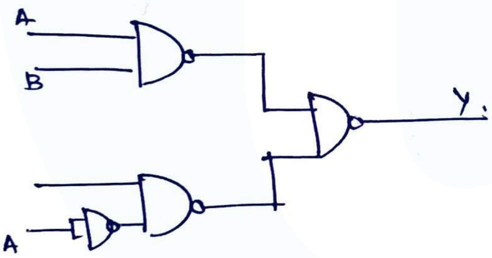
X-OR using NOR



Q:- Implement $Y = AB + \bar{A}C$ in minimum no. of NAND and NOR gates.

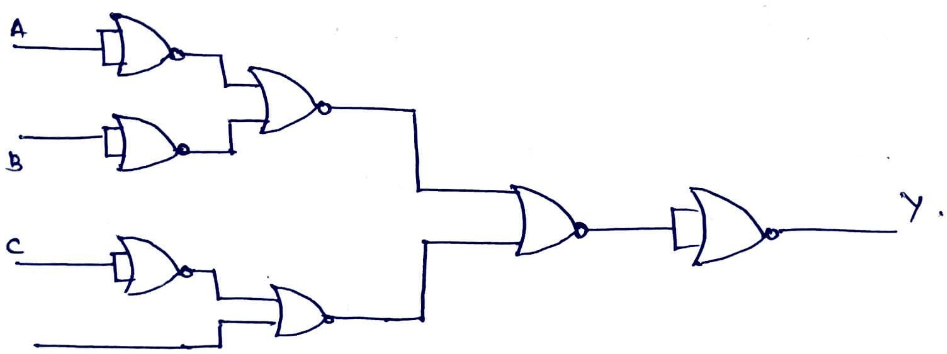
$$Y = AB + \bar{A}C.$$

$$= \overline{\overline{AB + \bar{A}C}} = \overline{\overline{AB} \cdot \overline{\bar{A}C}}.$$



$$Y = AB + \bar{A}C = \overline{\overline{AB} \cdot \overline{\bar{A}C}} = \overline{(\bar{A} + \bar{B}) \cdot (A + \bar{C})}$$

$$= \overline{\overline{\overline{(\bar{A} + \bar{B})} + \overline{(A + \bar{C})}}}$$



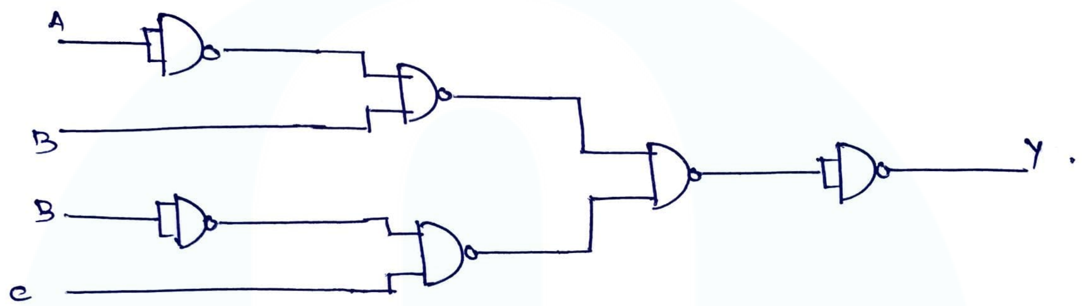
Q:- $Y = (A + \overline{B}) \cdot (B + \overline{C})$

$$Y = (A + \overline{B}) \cdot (B + \overline{C})$$

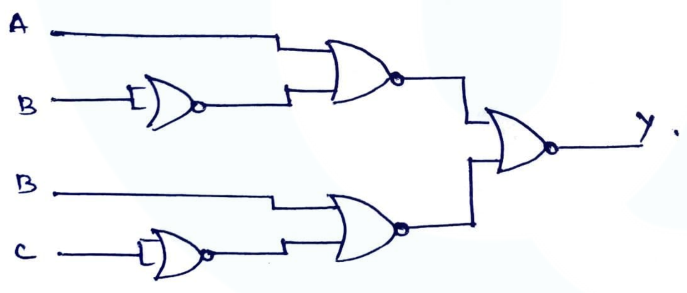
$$= \overline{\overline{(A + \overline{B}) \cdot (B + \overline{C})}} = \overline{\overline{A + \overline{B}} + \overline{(B + \overline{C})}}$$

$$= \overline{(\overline{A} \cdot B) + (\overline{B} \cdot C)}$$

$$= \overline{\overline{\overline{(\overline{A} \cdot B)} \cdot \overline{(\overline{B} \cdot C)}}}$$



$$Y = \overline{\left(\overline{A + \overline{B}} + \overline{B + \overline{C}} \right)}$$



Q:- $Y = ABC + A\overline{B}$

State & Prove De-Morgan's Theorem

$$\overline{A+B} = \overline{A} \cdot \overline{B}.$$

$$\overline{A \cdot B} = \overline{A} + \overline{B}.$$

A	B	$A+B$	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$	$A \cdot B$	$\overline{A \cdot B}$	$\overline{A} + \overline{B}$.
0	0	0	1	1	1	1	0	1	1
0	1	1	0	1	0	0	0	1	1
1	0	1	0	0	1	0	0	1	1
1	1	1	0	0	0	0	1	0	0.